A0: Warm-up: the five tools

Worked solutions

Training dynamics, a self-study course · written by Claude for Fan Pu

October 29, 2026

`reference numbers: solutions/results.json · figures: solutions/figures/ · equation numbers (n) refer to the handout`

Every derivation here is complete; read only as far as you need. "Measured" numbers come from running each check script with the reference implementations (`--solutions`) on CPU; your numbers should agree within the tolerances stated in each problem. Diff yours with `uv run python -m tdyn.results a0-prereqs/out/results.json a0-prereqs/solutions/results.json`. Equation numbers such as (2) refer to the handout; this document numbers its own equations as (S1), (S2), …

1 Problem (`flow-limit`): GD is forward Euler on gradient flow

(a) Forward Euler with velocity field $F(\theta) = -\nabla L(\theta)$ and time step h is $\theta_{k+1} = \theta_k + hF(\theta_k) = \theta_k - h\nabla L(\theta_k)$, which is GD (2) with $h = \eta$. Each step advances time by η, so θ_k approximates the flow at

$$t_k = k\eta. \tag{S1}$$

(b) Along the flow, $\dot\theta = -\nabla L(\theta)$ and, by the chain rule, $\ddot\theta = -\nabla^2 L(\theta)\dot\theta = \nabla^2 L(\theta)\nabla L(\theta)$. Taylor's theorem around $t = 0$ gives

$$\theta(\eta) = \theta_0 - \eta\nabla L(\theta_0) + \frac{\eta^2}{2}\nabla^2 L(\theta_0)\nabla L(\theta_0) + O(\eta^3), \tag{S2}$$

and one GD step is $\theta_1 = \theta_0 - \eta\nabla L(\theta_0)$, so

$$\theta(\eta) - \theta_1 = \frac{\eta^2}{2}\,\nabla^2 L(\theta_0)\nabla L(\theta_0) + O(\eta^3). \tag{S3}$$

The local error is $O(\eta^2)$. Reaching a fixed time T takes $K = T/\eta$ steps; if the local errors add up without being amplified, the global error is $K \cdot O(\eta^2) = O(\eta)$, so $p = 1$: GD is a first-order method. (More carefully, an error made at step j is carried forward by the flow and can grow by at most $e^{M(T-t_j)}$ with M the Lipschitz constant of ∇L; for a fixed T this is a constant, so the order stays 1. This is the standard Grönwall argument.)

(c) Expected. The slopes are 2 (one-step) and 1 (global). The leading term Equation (S3) even predicts the one-step error itself: at $(1.5, 0.5)$ the residual is $r = -0.25$, $\nabla L = (br, ar) = (-0.125, -0.375)$, $\nabla^2 L = \left(\begin{smallmatrix} 0.25 & 0.5 \\ 0.5 & 2.25 \end{smallmatrix}\right)$, so $\|\nabla^2 L \nabla L\| = \|(-0.21875, -0.90625)\| = 0.9323$ and the prediction is $0.466\,\eta^2$.

η	**one-step error**	$0.466\eta^2$ **from Equation (S3)**	**max error on** $[0, 10]$
0.1	4.37×10^{-3}	4.66×10^{-3}	8.13×10^{-3}
0.03	4.12×10^{-4}	4.20×10^{-4}	2.25×10^{-3}
0.01	4.63×10^{-5}	4.66×10^{-5}	7.35×10^{-4}
0.003	4.19×10^{-6}	4.19×10^{-6}	2.19×10^{-4}

Table 1: Measured errors of GD against the flow on the anchor from $(1.5, 0.5)$. Fitted log-log slopes: 1.983 (one-step) and 1.030 (max error).

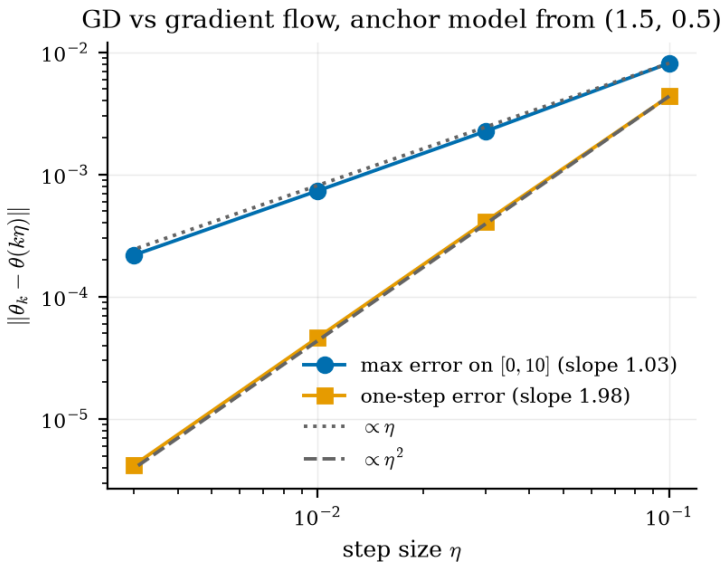


Figure 1: `flow_limit_error.png`: the two errors against η, with reference lines of slope 1 and 2.

(d) At $\eta = 0.9$ GD does *not* end at the flow's minimum. The flow goes monotonically to $(1.5538, 0.6436)$, the point of $ab = 1$ with $a^2 - b^2 = 2$ (the post's formula $a_\infty^2 = \left(c + \sqrt{c^2+4}\right)/2$ with $c = 2$). GD overshoots: ab crosses 1 at every one of the 60 steps, the imbalance $a^2 - b^2$ falls from 2 to 0.562, and GD settles at $(1.1488, 0.8704)$, a different and more balanced minimum. Why: the small-step limit needs η to be small *relative to the curvature*, i.e. $\eta\lambda_{\max} \ll 1$, not small in absolute terms. Near the flow's endpoint the sharpness is $a^2 + b^2 = \sqrt{c^2+4} = 2\sqrt{2} = 2.83$, so $\eta\lambda_{\max} = 2.55 > 2$: by Problem (`gd-stability`) GD cannot converge to that point at all, and it oscillates across the valley until it reaches minima flat enough to be stable (Problem (`hessian`) (c)). The post's §3.1 gives the factor by which each step changes the imbalance, $1 - \eta^2 r_k^2$, which is far from 1 while the oscillation is large.

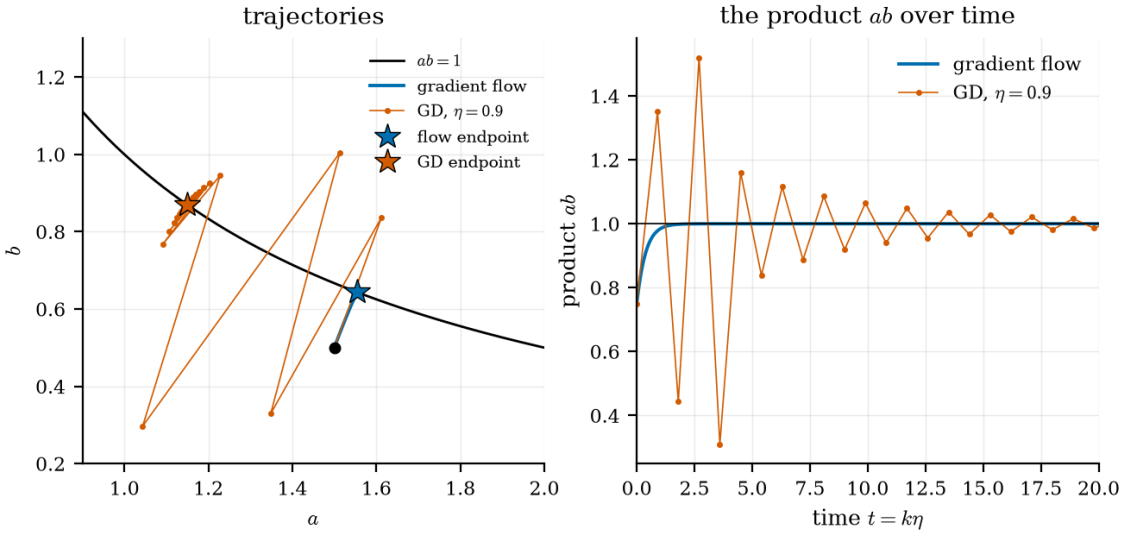


Figure 2: `flow_limit_eta09.png`: GD at $\eta = 0.9$ against the flow from $(1.5, 0.5)$.

2 Problem (`gd-stability`): Stability of GD on a quadratic

(a) $\nabla L(\theta) = \lambda\theta$, so $\theta_{k+1} = (1-\eta\lambda)\theta_k$ and

$$\theta_k = (1-\eta\lambda)^k \theta_0. \tag{S4}$$

With $q = 1-\eta\lambda$ (and $\eta\lambda > 0$): (i) monotone convergence for $0 < \eta\lambda < 1$ ($0 < q < 1$; at $\eta\lambda = 1$ exactly, $q = 0$ and GD lands on the minimum in one step); (ii) convergence with sign flips for $1 < \eta\lambda < 2$ ($-1 < q < 0$); (iii) at $\eta\lambda = 2$, $q = -1$ and $\theta_k = (-1)^k\theta_0$ bounces forever with constant amplitude; (iv) divergence for $\eta\lambda > 2$ ($q < -1$), with growing sign flips.

(b) With $H = Q\Lambda Q^\top$ and $z = Q^\top\theta$: $z_{k+1} = Q^\top(\theta_k - \eta Q\Lambda Q^\top\theta_k) = (I - \eta\Lambda)z_k$, so each coordinate obeys (a) with its own λ_i: $z_{i,k} = (1-\eta\lambda_i)^k z_{i,0}$. Since Q is orthogonal, $\|\theta_k\| = \|z_k\|$. (i) GD converges from every start iff $|1-\eta\lambda_i| < 1$ for all i, i.e. (all $\lambda_i > 0$)

$$\eta < \frac{2}{\lambda_{\max}}. \tag{S5}$$

(ii) For a generic start (every $z_{i,0} \neq 0$) the slowest-shrinking coordinate dominates, so

$$\rho(\eta) = \max_i |1-\eta\lambda_i| = \max(|1-\eta\lambda_{\min}|, |1-\eta\lambda_{\max}|). \tag{S6}$$

The first term decreases in η and the second increases once $\eta > 1/\lambda_{\max}$, so ρ is minimized where they cross: $1-\eta\lambda_{\min} = \eta\lambda_{\max} - 1$, giving $\eta^* = 2/(\lambda_{\min}+\lambda_{\max})$ and $\rho(\eta^*) = (\lambda_{\max}-\lambda_{\min})/(\lambda_{\max}+\lambda_{\min}) = (\kappa-1)/(\kappa+1)$ with $\kappa = \lambda_{\max}/\lambda_{\min}$ the *condition number*. (iii) For eigenvalues $\{1, 10\}$:

$$\eta_c = 0.200, \quad \eta^* = \frac{2}{11} = 0.182, \quad \rho(\eta^*) = \frac{9}{11} = 0.818. \tag{S7}$$

Below η^* the slow mode ($\lambda = 1$) limits the rate, above it the stiff mode ($\lambda = 10$) does; for $\eta > 1/\lambda_{\max} = 0.1$ the stiff mode already oscillates while GD still converges.

(c) Reference: `solutions/gd_solution.py`, a loop that copies `theta0`, applies (2) and stores every iterate.

(d) Expected. Measured on the grid: $\eta_c = 0.2000$, $\eta^* = 0.1820$, $\rho(\eta^*) = 0.8183$; and $\hat{\rho} = 0.99, 0.95, 0.90, 0.85, 0.90$ at $\eta = 0.01, 0.05, 0.1, 0.15, 0.19$, exactly $\max(|1-\eta|, |1-10\eta|)$.

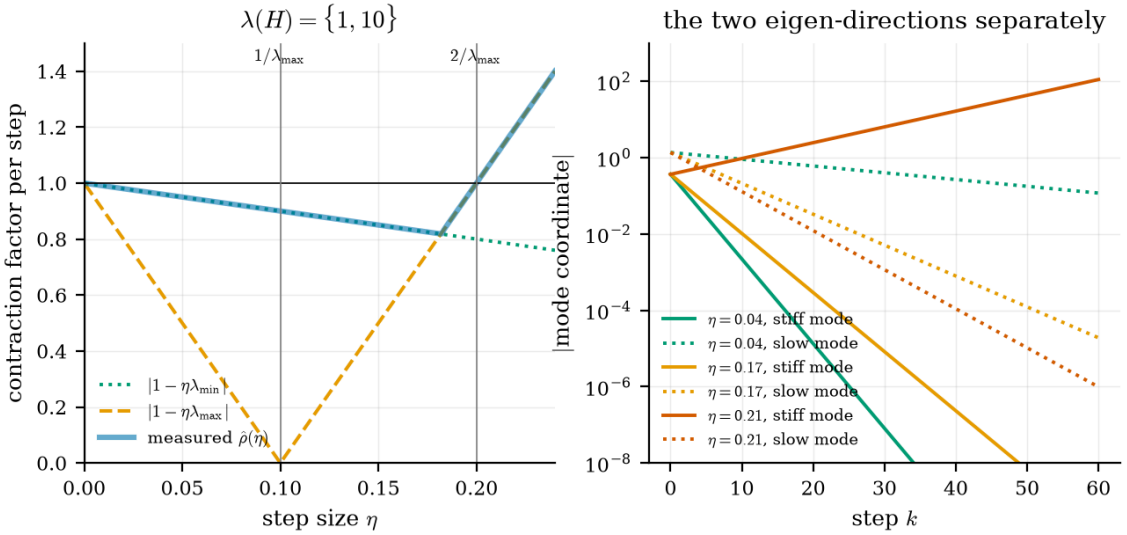


Figure 3: `gd_stability.png`. Left: measured contraction factor against the two mode factors; the measured curve is their maximum. Right: the two eigen-coordinates for $\eta = 0.04$ (both monotone), 0.17 (stiff mode oscillates and shrinks by 0.7 per step) and 0.21 (stiff mode grows by 1.1 per step).

3 Problem (`phase-portrait`): The anchor model's phase portrait

(a) With $r = ab - 1$: $\partial_a L = br$, $\partial_b L = ar$, so

$$\dot{a} = -b(ab-1), \quad \dot{b} = -a(ab-1). \tag{S8}$$

A fixed point needs $br = 0$ and $ar = 0$. If $r = 0$ both hold: every point of the hyperbola $ab = 1$ is a fixed point, with loss 0. If $r \neq 0$ then $a = b = 0$: the origin, with loss $1/2$. These are all.

(b) Writing $F = (-ab^2 + b, -a^2 b + a)$,

$$J(a,b) = \begin{pmatrix} -b^2 & 1-2ab \\ 1-2ab & -a^2 \end{pmatrix} = -\nabla^2 L(a,b). \tag{S9}$$

Origin: $J = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$, eigenvalues ± 1 with eigenvectors $(1,1)/\sqrt{2}$ (eigenvalue $+1$: trajectories leave along $a = b$) and $(1,-1)/\sqrt{2}$ (eigenvalue -1: they arrive along $a = -b$). A saddle. *On $ab = 1$:* at $(a, 1/a)$, $J = -\left(\begin{smallmatrix} 1/a^2 & 1 \\ 1 & a^2 \end{smallmatrix}\right)$, with $\det J = 1 - 1 = 0$ and $\operatorname{tr} J = -(a^2 + 1/a^2)$, so the eigenvalues are

$$\mu_1 = 0, \quad \mu_2 = -(a^2 + b^2) = -\left(a^2 + \frac{1}{a^2}\right). \tag{S10}$$

The eigenvector of $\mu_1 = 0$ is $(1, -1/a^2)$, which is $d/da\,(a, 1/a)$: the tangent to the curve $ab = 1$. The eigenvector of μ_2 is $(b, a) = \nabla(ab)$, the normal. So a perturbation off the curve decays at rate $a^2 + b^2$, and a perturbation along it neither grows nor decays, because it lands on another fixed point. By Table S1 the point is degenerate; the minima form a *line of neutrally stable fixed points*, and which one training reaches is decided by something other than the loss (here the conserved c; this is the post's implicit-bias question).

Point	**Eigenvalues of J**	**Label**
$(0,0)$	$-1, +1$	saddle
$(a, 1/a)$	$0, -(a^2 + 1/a^2)$	degenerate

Table S1: Linearization of the anchor's gradient flow.

(c) $d/dt(a^2 - b^2) = 2a\dot{a} - 2b\dot{b} = -2abr + 2abr = 0$. The origin has $c = 0$, so only initializations with $a_0^2 = b_0^2$, i.e. $a_0 = \pm b_0$, can reach it. On $a = b = s$ the flow is $\dot{s} = s(1 - s^2)$, which sends every $s \neq 0$ to ± 1. On $a = -b$, write $a = s$, $b = -s$: $\dot{a} = -b(ab - 1) = s(-s^2 - 1)$, so

$$\dot{s} = -s(1 + s^2), \tag{S11}$$

which decays to 0 from every start. So exactly the initializations on the line $a + b = 0$ end at the origin (the saddle's stable direction from (b)). A one-line proof of the colouring in the check: $d/dt(a + b) = -(a + b)(ab - 1)$, so $(a + b)(t) = (a_0 + b_0)\exp\left(-\int_0^t r\right)$ never changes sign. Starts with $a_0 + b_0 > 0$ end on the branch $a, b > 0$, those with $a_0 + b_0 < 0$ on $a, b < 0$.

(d) Reference: `solutions/fixed_points_solution.py`. It computes the eigenvalues with `np.linalg.eigvals`, returns `"degenerate"` if any has modulus below the tolerance, then separates complex pairs (spiral or center) from real pairs (node or saddle). The trace–determinant test of Example (`classify-by-hand`) is an equivalent route.

(e) Expected. Finite-difference eigenvalues: $(-1, 1)$ at the origin; $(-4.25, 0)$ at $a = 0.5$ and $a = 2$; $(-2, 0)$ at $a = \pm 1$, matching $-(a^2 + 1/a^2)$ to 10^{-10}. Labels: saddle, then degenerate four times. Of 156 initializations, 100% reach the branch given by the sign of $a_0 + b_0$; final loss below 10^{-27}; largest drift of c 5.6×10^{-8} (the RK4 error at step 0.005); a start on $a = -b$ ends 1.8×10^{-9} from the origin.

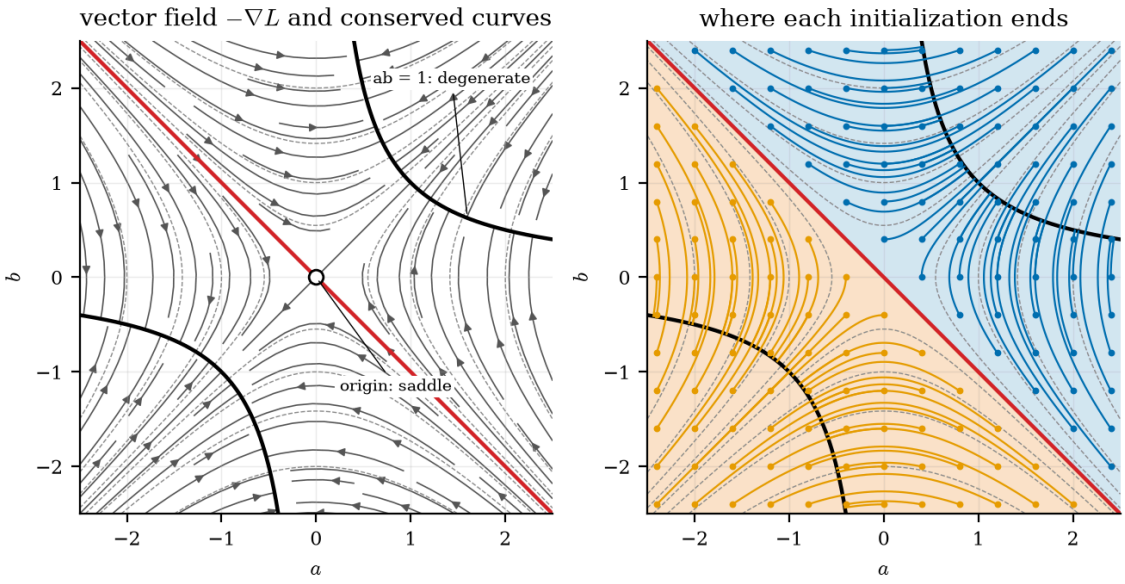


Figure 4: `phase_portrait.png`, the post's picture. Left: streamlines of $-\nabla L$, the conserved curves $a^2 - b^2 = c$ (dashed; the streamlines lie on them), $ab = 1$ (black), the separatrix $a = -b$ (red). Right: trajectories from a 13×13 grid, blue if they reach $a, b > 0$, orange if $a, b < 0$, on the predicted regions $a + b > 0$ and $a + b < 0$.

4 Problem (`hessian`): Sharpness and trace from Hessian-vector products

(a) Reference: `solutions/hessian_solution.py`. `hvp` makes `params` a fresh leaf, takes `g = grad(loss, p, create_graph=True)`, then `grad(g @ v, p)`. `power_iteration` draws v_0 from `torch.Generator().manual_seed(seed)`, normalizes, applies (5) `iters` times and returns the Rayleigh quotient of the last vector. `hutchinson_trace` averages $z^\top H z$ over Rademacher z (`randint(0, 2) * 2 - 1`). Two remarks. With Rademacher probes $z^\top H z = \sum_i H_{ii} + \sum_{i \neq j} z_i z_j H_{ij}$, so the variance is $2 \sum_{i \neq j} H_{ij}^2$ and a diagonal H is estimated exactly from one probe (that is what `test_hutchinson_trace_diagonal_is_exact` checks); Gaussian probes add a $2 \sum_i H_{ii}^2$ term. Power iteration's rate is $|\lambda_2/\lambda_1|$ per step: 0.74 for the test MLP (2.457/3.319), so 100 steps reach machine precision.

(b) From Problem (`phase-portrait`) (b), $\nabla^2 L = -J$ has eigenvalues 0 and $a^2 + b^2$ on the manifold, so

$$\lambda_{\max}(\nabla^2 L) = a^2 + \frac{1}{a^2}, \quad \operatorname{tr} \nabla^2 L = a^2 + \frac{1}{a^2}. \tag{S12}$$

Only the nonzero eigenvalue matters for linear stability: the zero direction is neutral to first order, since neighbouring points of the manifold are also minima. Stability needs $a^2 + 1/a^2 < 2/\eta$. With $u = a^2$ this is $u^2 - (2/\eta)u + 1 < 0$, i.e. $u \in \left(1/\eta - \sqrt{1/\eta^2 - 1}, 1/\eta + \sqrt{1/\eta^2 - 1}\right)$. For $\eta = 0.5$: $a^2 \in \left(2 - \sqrt{3}, 2 + \sqrt{3}\right)$, so

$$a \in (0.5176, 1.9319). \tag{S13}$$

For $\eta = 1$: $a^2 + 1/a^2 \geq 2$ with equality only at $a = 1$, so no minimum is strictly stable; $(1, 1)$ is marginal ($\eta \lambda_{\max} = 2$, case (iii) of Problem (`gd-stability`)). The largest step size with a stable minimum is therefore $\eta < 1$, approached by the balanced minimum, which A3 develops.

(c) Expected. Sharpness along $ab = 1$ matches $a^2 + 1/a^2$ to 4×10^{-16} relative; the trace estimates are within 0.010 (400 probes: $z^\top H z = a^2 + b^2 \pm 2$, so the standard error is 0.1). Measured stable set for $\eta =$

0.5 on the grid: $a \in [0.527, 1.899]$, inside $(0.518, 1.932)$ up to the 4.7% grid spacing. MLP (193 parameters): power iteration 3.319157 against the exact 3.319157; Hutchinson with 200 probes 10.12 against the exact trace 9.74 (standard error 0.55). The two stars: the flow's endpoint $(1.554, 0.644)$ has sharpness $2\sqrt{2} = 2.828 > 2/0.9 = 2.222$, so it is linearly *unstable* for GD at $\eta = 0.9$, and GD could never settle there. The GD endpoint $(1.149, 0.870)$ has sharpness $2.077 < 2.222$, inside the stable set $a \in (0.792, 1.263)$ for $\eta = 0.9$. GD at a large step size can only stop at minima flatter than $2/\eta$: a finite step size biases GD toward flat, balanced minima. This is the picture A3 starts from.

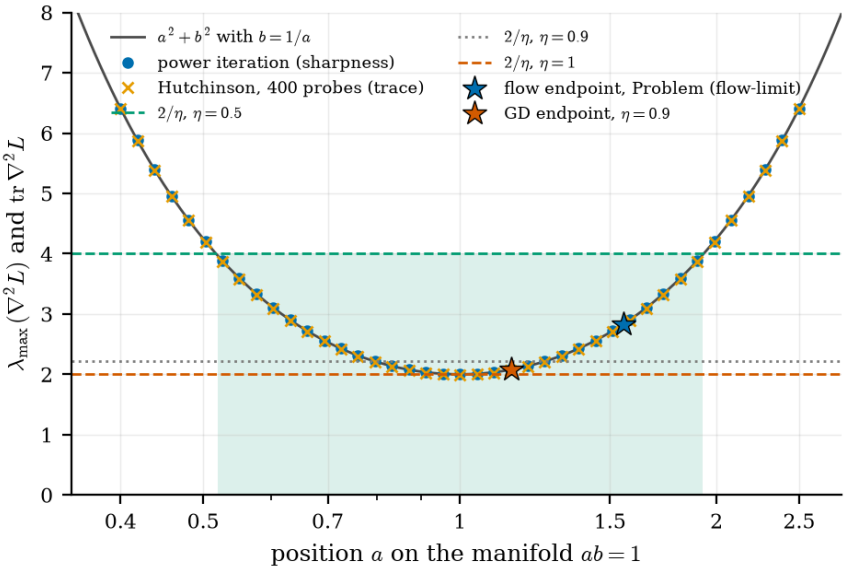


Figure 5: `sharpness.png`: sharpness (power iteration) and trace (Hutchinson) along $ab = 1$, the thresholds $2/\eta$, the stable band for $\eta = 0.5$ (shaded) and the two endpoints of Problem (`flow-limit`).

5 Problem (`modes`): Gradient flow of linear regression, mode by mode

(a) $L(W) = \frac{1}{2}\mathbb{E}[(Wx - y)^\top (Wx - y)]$. Perturbing $W \to W + \Delta$ changes L by $\mathbb{E}[(Wx - y)^\top \Delta x] = \operatorname{tr}(\Delta^\top \mathbb{E}[(Wx - y)x^\top])$ to first order, so

$$\nabla_W L = \mathbb{E}[(Wx - y)x^\top] = W\Sigma_{xx} - \Sigma_{yx}, \tag{S14}$$

and gradient flow is $\dot{W} = \Sigma_{yx} - W\Sigma_{xx}$, equation (8). Setting the gradient to zero: $W^\star = \Sigma_{yx}\Sigma_{xx}^{-1}$, the least-squares solution.

(b) Let $E = W - W^\star$. Since $W^\star \Sigma_{xx} = \Sigma_{yx}$, $\dot{E} = -E\Sigma_{xx}$. Rotate: $\tilde{E} = EU$ with $\Sigma_{xx} = U\Lambda U^\top$; then $\dot{\tilde{E}} = -EU\Lambda U^\top U = -\tilde{E}\Lambda$. Column i of $\tilde{E}$, the error of mode i, obeys $\dot{\tilde{e}}_i = -\lambda_i \tilde{e}_i$ by itself, so $\tilde{e}_i(t) = e^{-\lambda_i t}\tilde{e}_i(0)$. Rotating back,

$$W(t) = W^\star + (W_0 - W^\star)Ue^{-\Lambda t}U^\top = W^\star + (W_0 - W^\star)e^{-\Sigma_{xx} t}. \tag{S15}$$

Mode i shrinks by a factor e in time $1/\lambda_i$, so the smallest eigenvalue of the input covariance sets the training time, and an ill-conditioned Σ_{xx} means slow learning (the continuous-time face of Problem (`gd-stability`)'s condition number).

(c) Reference: `solutions/modes_solution.py`: `eigh` of Σ_{xx}, $W^\star$ by a linear solve, then the formula above.

(d) Expected. Fitted decay rates $0.1000, 0.3162, 1.0000, 3.1623, 10.000$, equal to the eigenvalues to a relative 10^{-7}; closed form within 2×10^{-8} of RK4 (step 0.005).

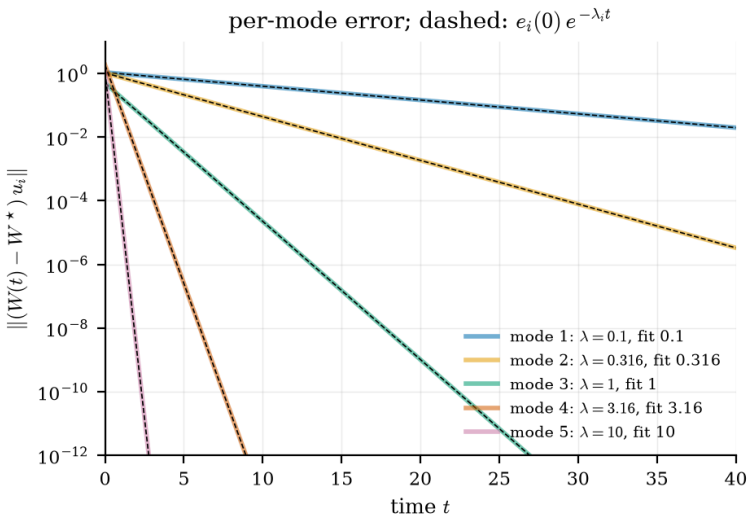


Figure 6: `modes.png`: per-mode error $\|(W(t) - W^\star)u_i\|$; each is a straight line of slope $-\lambda_i$ on the log scale.

(e) With $\Sigma_{xx} = I$ and $W_0 = 0$: $W^\star = \Sigma_{yx}$ and $W(t) = (1 - e^{-t})\Sigma_{yx}$. If $\Sigma_{yx} = USV^\top$, then $W(t) = U[(1 - e^{-t})S]V^\top$: the singular vectors are those of Σ_{yx} from the first instant, and every singular value grows as $s_\alpha(1 - e^{-t})$, all at the same rate, with no plateau. Saxe et al. show that with two layers each singular value instead follows an S-shaped curve whose timing scales like $1/s_\alpha$, so strong modes are learned first (A1).

6 Problem (ou): The Ornstein–Uhlenbeck process

(a) Taking expectations in (10) with $\mu(X) = -\theta X$, and using $\mathbb{E}[\xi_k] = 0$: $m_{k+1} = (1 - \theta\Delta t)m_k$, so $(m_{k+1} - m_k)/\Delta t = -\theta m_k$, which tends to $\dot{m} = -\theta m$. Hence

$$m(t) = x_0 e^{-\theta t}. \tag{S16}$$

(Directly: $m_k = x_0(1 - \theta\Delta t)^k \to x_0 e^{-\theta t}$ with $k\Delta t = t$.)

(b) Subtract the mean recursion: $X_{k+1} - m_{k+1} = (1 - \theta\Delta t)(X_k - m_k) + \sigma\sqrt{\Delta t}\xi_k$, and ξ_k is independent of X_k, so variances add:

$$v_{k+1} = (1 - \theta\Delta t)^2 v_k + \sigma^2\Delta t. \tag{S17}$$

Expanding, $(v_{k+1} - v_k)/\Delta t = -2\theta v_k + \sigma^2 + \theta^2\Delta t\, v_k$; as $\Delta t \to 0$, $\dot{v} = -2\theta v + \sigma^2$ with $v(0) = 0$, so

$$v(t) = \frac{\sigma^2}{2\theta}\big(1 - e^{-2\theta t}\big), \quad v_\infty = \frac{\sigma^2}{2\theta}. \tag{S18}$$

The noise term is $\sigma^2\Delta t$, first order in Δt, only because the increment is scaled by $\sqrt{\Delta t}$; with Δt instead, the noise would vanish in the limit.

(c) Reference: `solutions/sde_solution.py`, a loop over steps, vectorized over paths, with `rng.standard_normal(n_paths)` per step.

(d) Expected. At $t = 1$: sample mean 0.7396 (formula 0.7358; standard error 0.0046), variance 0.1096 (formula 0.1081). At $t = 8$: variance 0.1290 against $v_\infty = 0.125$, within 1.6 standard errors (0.0025) plus the Euler–Maruyama bias +0.5% from (e).

(e) Prediction to compare with: the stationary variance grows with Δt and blows up as $\theta\Delta t \to 2$. The fixed point of the exact recursion Equation (S17) is

$$v_\infty^{\mathrm{EM}} = \frac{\sigma^2 \Delta t}{1-(1-\theta\Delta t)^2} = \frac{\sigma^2}{\theta(2-\theta\Delta t)}, \tag{S19}$$

which equals $\sigma^2/(2\theta)$ as $\Delta t \to 0$ and diverges at $\theta\Delta t = 2$. For $\theta\Delta t > 2$ there is no stationary distribution at all: the mean recursion has $|1-\theta\Delta t| > 1$. Measured with $\theta = 1$, $\sigma = 0.5$: $0.123, 0.132, 0.146, 0.178, 0.249, 0.416, 0.833, 2.498$ at $\theta\Delta t = 0.01, 0.1, 0.3, 0.6, 1.0, 1.4, 1.7, 1.9$, against the formula's $0.126, 0.132, 0.147, 0.179, 0.250, 0.417, 0.833, 2.500$.

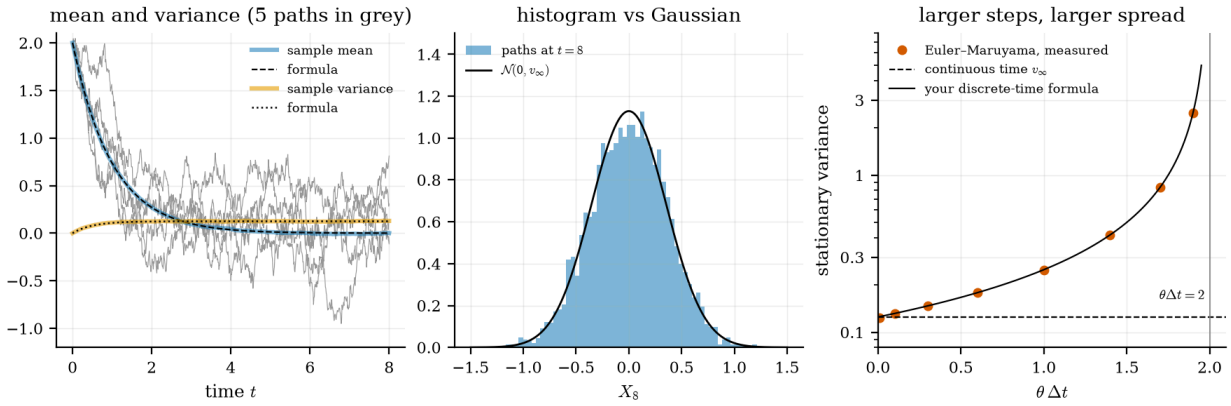


Figure 7: `ou.png`. Left: sample mean and variance against the formulas. Middle: the paths at $t = 8$ against $\mathcal{N}(0, v_\infty)$. Right: Euler–Maruyama's stationary variance against $\theta\Delta t$, with the continuous-time value and the exact discrete-time formula.

(f) Let $X_k = \theta_k - \theta^*$. SGD is $X_{k+1} = X_k - \eta(\lambda X_k + \xi_k) = X_k - \lambda X_k \eta - \eta\xi_k$. Compare with (10) for $\mu(X) = -\theta_{\mathrm{OU}} X$: the drift matches with $\theta_{\mathrm{OU}} = \lambda$ and $\Delta t = \eta$, and the noise matches when $\sigma\sqrt{\Delta t}\,\mathcal{N}(0,1)$ has the law of $-\eta\xi_k$, i.e. $\sigma\sqrt{\eta} = \eta s$, so $\sigma = s\sqrt{\eta}$ (if ξ_k is Gaussian; otherwise the match is in mean and variance). Plugging into (e):

$$\mathrm{Var}[\theta_k - \theta^*] \to \frac{\eta s^2}{\lambda(2-\eta\lambda)} \approx \frac{\eta s^2}{2\lambda} \quad (\eta\lambda \ll 1). \tag{S20}$$

The spread grows linearly with the step size and the gradient-noise variance and shrinks with the curvature: sharp directions hold the iterate close, flat directions let it wander. (The expected excess loss $\lambda/2 \cdot \mathrm{Var} \approx \eta s^2/4$ does not depend on λ at all.) The divergence at $\eta\lambda = 2$ is the GD stability threshold of Problem (`gd-stability`): noise does not move the threshold, it only makes the approach to it visible as a growing spread. A4 runs this on the anchor, where the minimum is a manifold and the curvature $a^2 + b^2$ changes along it, which turns the fluctuation into a drift.

7 Problem (`framing`): reading notes

There is no single right answer; here is what a good answer covers, with page pointers. (a) The anchor is a *solvable setting*: Simon et al. list deep linear networks $f(x;\theta) = W_L \ldots W_1 x$ with $L \geq 2$ as their first example of an analytically solvable setting (§2.1, eq. (1), p. 6), and the anchor is the scalar $L = 2$ case. From it the course reaches *insightful limits* (small initialization and lazy vs rich learning, Table 1 row 2.2; A5), *simple empirical laws* (the edge of stability, row 2.3; A3) and *theories of hyperparameters* (step size as sharpness regularization, µP, row 2.4; A3 and A7). (b) For the anchor under gradient flow, the conserved $c = a_0^2 - b_0^2$ is an "early signal" that predicts the outcome exactly: measured at any time, it gives the endpoint $a_\infty^2 = \left(c + \sqrt{c^2+4}\right)/2$, the sharpness there, $\sqrt{c^2+4}$, and hence (Problem (`hessian`)) whether GD at step η can settle there. Under finite steps or noise c drifts, and predicting from early training means predicting that drift (A1, A3, A4). This is the first of Biderman et al.'s capabilities,

predicting training (§2.4); *intervening* would be, for example, lowering η when the measured sharpness approaches $2/\eta$. (c) Personal. Candidates worth arguing about: whether coarse statistics will suffice for safety-relevant behaviour (Simon et al. §1.2), and whether "predict, intervene, design" is really a hierarchy rather than three partly independent skills (Biderman et al. §2.4).